
Outlier Migration in Long Decode Traces

Anonymous authors
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Abstract

This draft reports a conservative Phase 0 result for OutlierMigrate, a candidate positive-method branch studying whether high-magnitude activation channels remain stationary during long decode. The preregistered Phase 0 gate passes on Granite-4.0-H-Tiny: the fraction of top-1% position-100 channels whose rank changes by more than two channel positions by decode position 10000 is 0.8178385416666667 with bootstrap 95% CI [0.797265625, 0.8368489583333334]. This is not camera-ready, is not a full positive-method claim, and does not provide cross-model evidence. Phase 1 scale validation is required before the work can support submission-level claims.

1 Introduction

OutlierMigrate asks whether the channels carrying high-magnitude decode-time activations are fixed or whether they move during long reasoning traces. Static outliers would support static channel-protection recipes. Dynamic outliers would motivate migration-aware protection or routing methods, but only after larger validation shows that the effect is stable and useful.

The current paper status is deliberately limited. Phase 0 is a screen on one model family, one model size, and 12 deterministic AIME-2025 reasoning traces. It supports a dynamic-outlier observation under the preregistered metric. It does not support a camera-ready method claim, an end-to-end efficiency claim, or any cross-model claim.

2 Method

The Phase 0 procedure follows the frozen preregistration at `experimental/outlier_migrate/phase0/preregister_om_phase0.md`. The model is `ibm-granite/granite-4.0-h-tiny` at HuggingFace snapshot commit `791e0d3d28c86e106c9b6e0b4cecdce0375b6124`. The prompt set contains 12 deterministically selected AIME-2025 traces, indices 0–11.

For every transformer layer, the runner records per-channel activation magnitudes at decode positions {100, 500, 1000, 5000, 10000}. It selects the top 1% channels by mean magnitude at position 100 and computes the fraction whose rank at position 10000 differs by more than two channel positions. The reported gate metric is the trace-bootstrap aggregate over layers.

3 Experiments

The Phase 0 result packet is `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z`. The checker output is `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/checker_result.json`, the metrics file is `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/metrics.json`, and artifact completeness is recorded in `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/artifact_check.json`.

The preregistered dynamic-outlier pass rule requires migration fraction at least 0.05 and bootstrap CI lower bound greater than 0.05. The static pass rule and ambiguous-effect kill rule are not used because the dynamic pass rule is satisfied.

Quantity	Value
Decision	PASS_OM_PHASE0_DECODE_TIME_MIGRATION
Migration fraction	0.8178385416666667
Bootstrap 95% CI	[0.797265625, 0.8368489583333334]
Dynamic threshold	migration fraction ≥ 0.05 and CI lower > 0.05
Model snapshot commit	791e0d3d28c86e106c9b6e0b4cecdde0375b6124
Layer count / hidden size	40 / 1536
Trace count	12

Table 1: Phase 0 gate result from the checker and metrics artifacts cited in Section 3. The result clears the preregistered dynamic threshold because $0.8178385416666667 \geq 0.05$ and the CI lower bound 0.797265625 is greater than 0.05.

4 Results

The checker returned PASS_OM_PHASE0_DECODE_TIME_MIGRATION. The exact checker reason was that the point estimate 0.81783854 is at least 0.05 and the CI lower bound 0.79726562 is greater than 0.05.

5 Discussion

The Phase 0 result weakens the static-outlier hypothesis on this exact surface: under the preregistered rank-migration metric, most position-100 top-1% channels do not remain rank-stationary through position 10000. This keeps OutlierMigrate alive as a candidate branch. It does not establish that migration-aware intervention improves model quality, latency, memory, or robustness.

The highest-priority next branch is Phase 1 scale validation. Phase 1 must test whether the same observation survives a larger frozen surface before any method paper can claim a reusable mechanism.

6 Limitations

This draft is not camera-ready. It reports one Phase 0 screen on one model snapshot, one prompt source, and 12 traces. There is no cross-model evidence, no same-family scale validation, no cross-family falsification pair, no seed repeat beyond the recorded bootstrap procedure, and no intervention showing that migration-aware handling improves an application metric. The result may also depend on the chosen decode positions, top-channel fraction, and rank delta threshold, all of which were fixed before analysis and should not be retuned on this packet.

7 Reproducibility Statement

The result packet is experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z. It includes environment.json, model_provenance.json, prompt_manifest.json, command_metadata.json, random_seed.json, metrics.json, checker_result.json, and artifact_check.json. The model snapshot commit is 791e0d3d28c86e106c9b6e0b4cecdde0375b6124. The recorded command was:

```
experimental/shared/run_phase0_branch.py --branch outlier_migrate
```

The checker decision and exact Phase 0 numbers in this draft are taken from the result directory above.

8 Ethics Statement

This is an activation-analysis systems study on benchmark prompts. The current artifact does not make deployment, user-data, safety, or capability-improvement claims. The main ethical risk is overclaiming a Phase 0 screen as a validated positive method; this draft explicitly rejects that interpretation until Phase 1 scale validation passes.